

IQL

Environment	ReBRAC-tuned	CORL-tuned + MPI	
	Baseline	π_1	π_2
HalfCheetah-medium	[49.8, 50.2] 50.0	[53.5, 54.0] 53.8	[55.3, 56.6] 55.9
HalfCheetah-medium-replay	[38.5, 45.7] 42.1	[49.3, 50.3] 49.8	[51.4, 53.2] 52.3
HalfCheetah-medium-expert	[89.9, 95.5] 92.7	[87.3, 94.7] 91.0	[82.7, 94.4] 88.6
Hopper-medium	[61.0, 69.4] 65.2	[81.1, 99.9] 90.5	[97.9, 104.9] 101.4
Hopper-medium-replay	[76.4, 102.8] 89.6	[97.1, 103.9] 100.5	[102.6, 103.5] 103.0
Hopper-medium-expert	[55.8, 115.2] 85.5	[95.3, 117.0] 106.1	[96.5, 118.2] 107.4
Walker2d-medium	[77.3, 84.1] 80.7	[59.8, 90.1] 74.9	[81.9, 88.6] 85.3
Walker2d-medium-replay	[66.1, 84.7] 75.4	[78.4, 85.2] 81.8	[78.9, 90.0] 84.5
Walker2d-medium-expert	[111.6, 112.6] 112.1	[111.9, 113.4] 112.6	[112.6, 115.0] 113.8
<i>Average</i>	77.0	84.6	88.0

Table 1: IQL: ReBRAC-tuned baseline vs. CORL-tuned MPI policies. Scores are D4RL-normalized returns; brackets denote [mean – std, mean + std].

ReBRAC

Environment	ReBRAC-tuned	CORL-tuned + MPI	
	Baseline	π_1	π_2
HalfCheetah-medium	[64.6, 66.6] 65.6	[64.4, 67.4] 65.9	[64.4, 67.6] 66.0
HalfCheetah-medium-replay	[50.2, 51.8] 51.0	[49.8, 51.7] 50.7	[49.5, 50.5] 50.0
HalfCheetah-medium-expert	[95.9, 106.3] 101.1	[92.8, 101.5] 97.1	[95.0, 99.1] 97.1
Hopper-medium	[101.0, 103.0] 102.0	[99.6, 102.5] 101.1	[98.9, 103.5] 101.2
Hopper-medium-replay	[92.8, 103.4] 98.1	[100.4, 101.0] 100.7	[98.0, 101.1] 99.5
Hopper-medium-expert	[100.6, 113.4] 107.0	[106.9, 112.4] 109.7	[99.3, 115.2] 107.2
Walker2d-medium	[78.9, 86.1] 82.5	[84.5, 86.6] 85.6	[86.0, 87.9] 87.0
Walker2d-medium-replay	[69.4, 85.2] 77.3	[81.3, 91.4] 86.3	[84.6, 92.1] 88.4
Walker2d-medium-expert	[111.3, 111.9] 111.6	[111.3, 112.9] 112.1	[113.1, 113.6] 113.4
<i>Average</i>	88.5	89.9	90.0

Table 2: ReBRAC: ReBRAC-tuned baseline vs. CORL-tuned MPI policies. Scores are D4RL-normalized returns; brackets denote [mean – std, mean + std].

TD3+BC

Environment	ReBRAC-tuned	CORL-tuned + MPI	
	Baseline	π_1	π_2
HalfCheetah-medium	[53.8, 55.6] 54.7	[54.1, 54.8] 54.5	[57.2, 58.2] 57.7
HalfCheetah-medium-replay	[43.9, 46.1] 45.0	[48.9, 50.6] 49.8	[50.9, 53.0] 52.0
HalfCheetah-medium-expert	[83.5, 94.7] 89.1	[86.2, 93.6] 89.9	[65.9, 82.9] 74.4
Hopper-medium	[53.3, 68.5] 60.9	[84.0, 98.9] 91.4	[99.1, 102.1] 100.6
Hopper-medium-replay	[23.4, 86.8] 55.1	[97.8, 100.8] 99.3	[97.5, 100.8] 99.1
Hopper-medium-expert	[77.3, 98.3] 87.8	[84.5, 109.1] 96.8	[97.2, 113.7] 105.5
Walker2d-medium	[74.8, 80.6] 77.7	[77.8, 90.8] 84.3	[87.3, 91.5] 89.4
Walker2d-medium-replay	[48.8, 87.2] 68.0	[83.1, 88.2] 85.7	[87.5, 91.3] 89.4
Walker2d-medium-expert	[109.8, 111.0] 110.4	[111.6, 112.8] 112.2	[112.1, 115.1] 113.6
<i>Average</i>	72.1	84.9	86.9

Table 3: TD3+BC: ReBRAC-tuned baseline vs. CORL-tuned MPI policies. Scores are D4RL-normalized returns; brackets denote [mean $-$ std, mean $+$ std].